Dangerzone: making documents safe

Our next release needs **YOU**!



What is Dangerzone?

A tool to render any document into a harmless document.

How it works:

1. Open the document in a hardened sandbox
2. Take a picture of each page
3. Construct a PDF from those images
4. Optionally, perform character recognition (OCR) on top

The sandbox

The sandbox is where potentially dangerous documents are processed safely.

- **A container** - Isolated environment
- **Running LibreOffice and a PDF reader** - Handles various document formats
- **Offline with restrictions** - No network access, limited capabilities

This isolation protects your system from malicious documents.

Problem #1

Software inside the container needs to be kept up to date

CVEs happen frequently:

- LibreOffice
- Debian
- Large attack surface

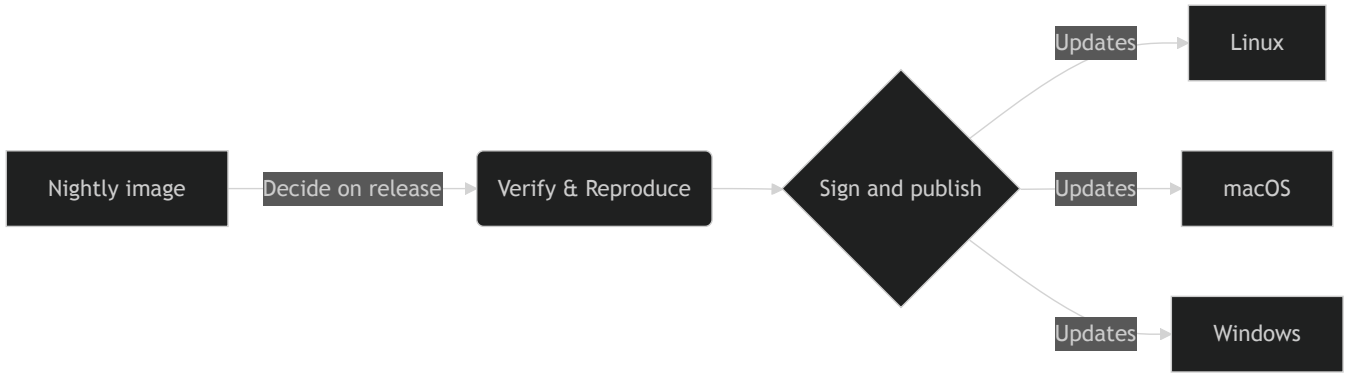
Dangerzone releases take significant work:

- We can't issue a new release for every container update.
- Users tend to lag behind

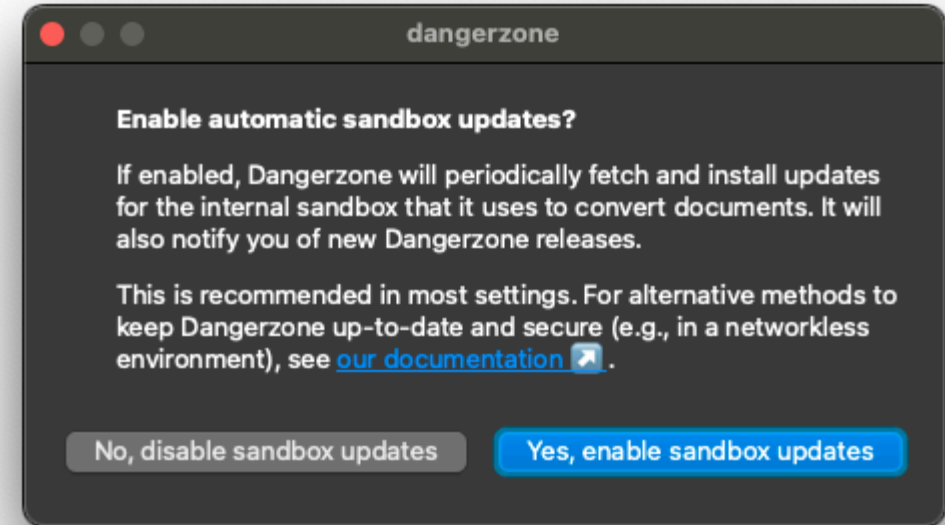


Solution #1: Auto-updating sandbox

Independent container updates that work automatically.



- Avoiding supply-chain attacks
- **Signed with FPF-owned keys** - You can trust the source
- **Reproducible builds** - Anyone can verify the container matches the source
- **Auditable** - signatures are uploaded to an external log

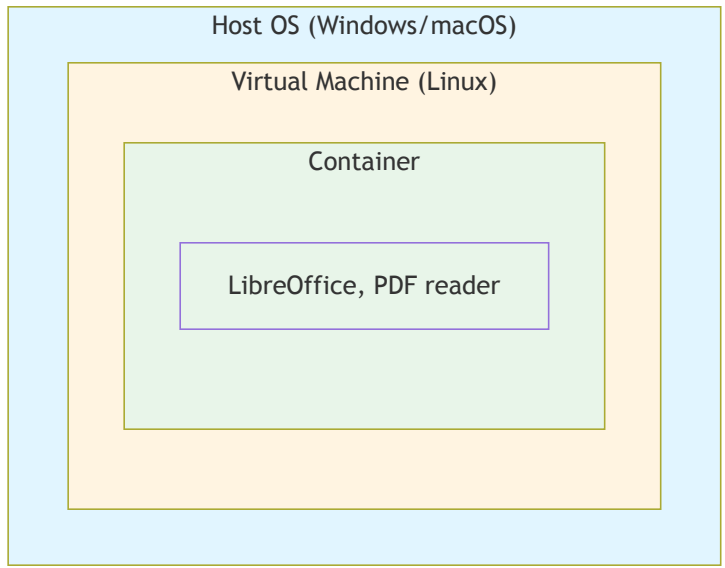

dangerzone
Enable automatic sandbox updates?
If enabled, Dangerzone will periodically fetch and install updates for the internal sandbox that it uses to convert documents. It will also notify you of new Dangerzone releases.
This is recommended in most settings. For alternative methods to keep Dangerzone up-to-date and secure (e.g., in a networkless environment), see our documentation ↗.
No, disable sandbox updates
Yes, enable sandbox updates

Some context

Windows and macOS need a VM

Containers rely on Linux kernel features, so on non-Linux systems:

- A lightweight VM provides the Linux kernel
- The container runs inside this VM
- This adds an extra layer between the container and your OS



Problem #2

The sandbox needs Docker Desktop (on macOS and Windows)

This creates issues:

- Another installation step for users
- Needs to be kept up to date to avoid exploits
- Licensing concerns (Docker Desktop is not open source)

And also…

Docker is its own beast:

```
$ docker desktop start
Docker Desktop is already running
$ docker desktop status
Could not retrieve status. Is Docker Desktop running?
You can start Docker Desktop by running 'docker desktop start'.
```

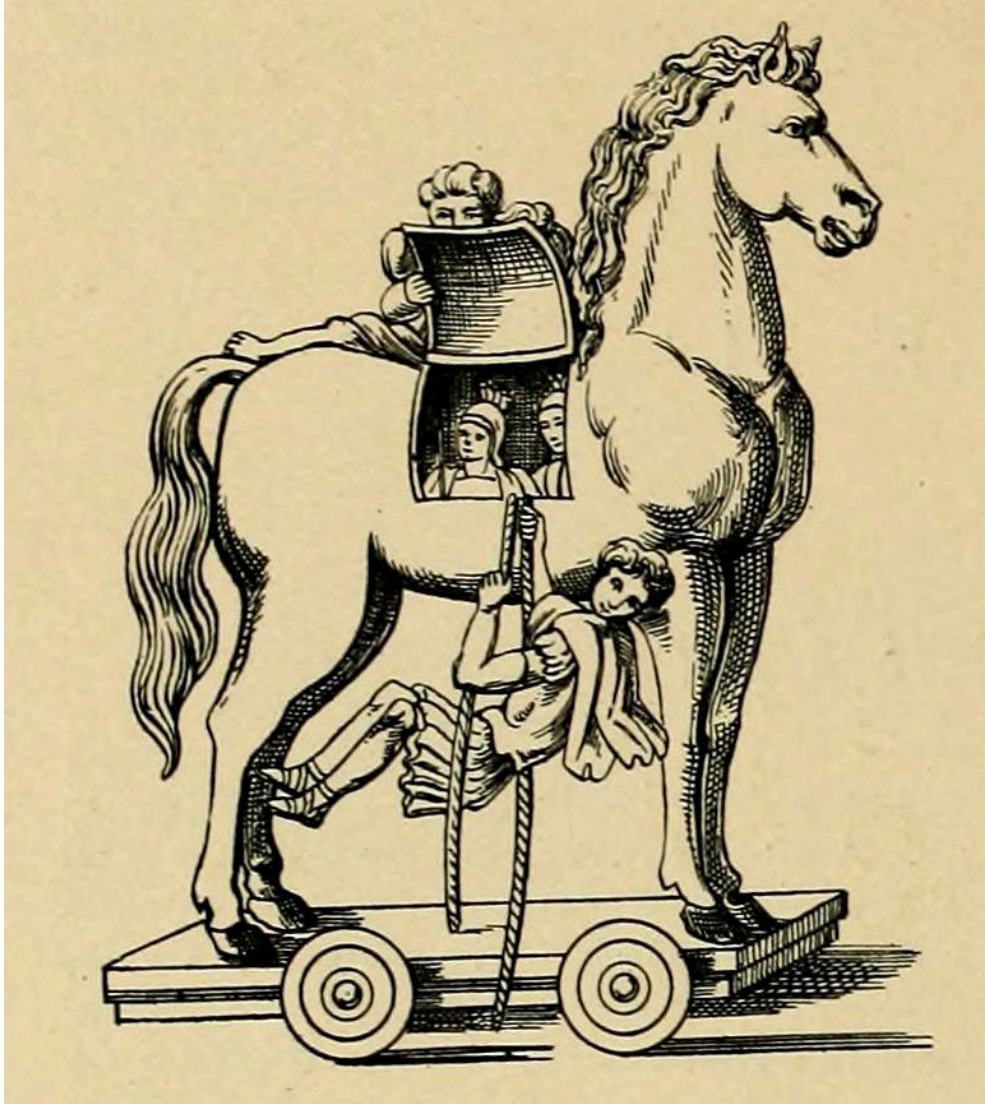
- We’ve had two emergency releases in the past due to Docker
- "It asked for it"

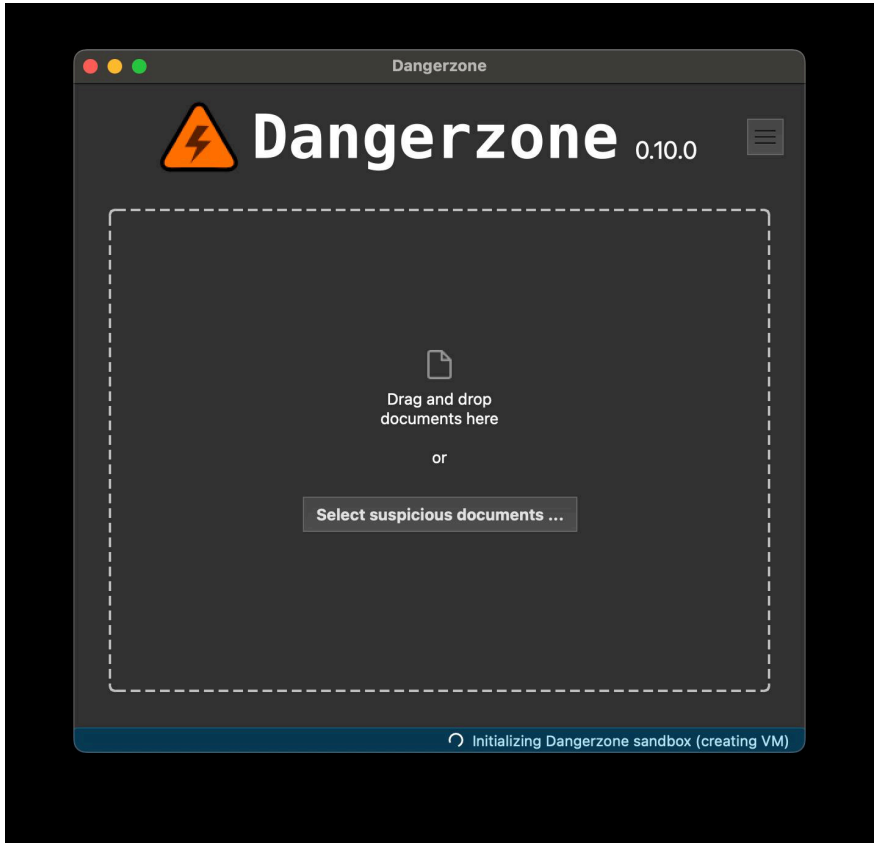
Solution #2: Embedded Podman

We're embedding Podman directly in our macOS and Windows builds!

Benefits:

- No separate Docker Desktop installation needed
- Simpler user experience
- No licensing issues
- Full control over the container runtime




Dangerzone
Dangerzone 0.10.0
Drag and drop
documents here
or
Select suspicious documents ...
Initializing Dangerzone sandbox (creating VM)

Help us test!

We need your help testing these new features.

- Especially on Windows and macOS
  - where we embed Podman handle the VMs etc.
- Embedding Podman results in quite large packages:
  - `.dmg` is 1.8Gib
  - `.msi` is 1.2Gib
- → https://github.com/freedomofpress/dangerzone/releases/tag/v0.10.0-rc1



Sources:

- Cheval de Troie d’après le Virgile du Vatican.jpg
- An aerial view of the San Andreas Fault in the Carrizo Plain, Central California
- Terry Johnston from Grand Rapids, USA, Creative Commons Attribution 2.0